

Gagan Jattenahalli Harishkumar

Personal website
jhgagan9@gmail.com
+39 3208434195

PROFILE

Telecommunications Engineer specializing in **semantic communication**, **physical-layer wireless simulation**, and **signal processing**. Experience with **6G-enabling technologies**: feature-based compression, gRPC distributed systems, embedded/edge AI. Extending current research toward **satellite-to-vehicle (NTN-V2X)** communication. 7 peer-reviewed publications (Scientific Data, IEEE Access, IEEE conferences).

EDUCATION

M.Sc. Telecommunication Engineering

Politecnico di Milano

Track: Signals and Data Analysis

CGPA: 25.79/30

Thesis: Semantic Communication Framework and Simulation with Sionna

Expected Graduation: October 2026

B.Tech. Electrical & Electronics Engineering

Manipal Institute of Technology

Specialization: Data Science and Analytics

CGPA: 8.17/10

Thesis: Raspberry-Pi Based Fundus Imaging Device

2017 - 2021

SKILLS & INTERESTS

Wireless & Communication Systems: Semantic Communication, Physical-Layer Simulation (Sionna), 6G, Non-Terrestrial Networks (NTN), V2X, Channel Modeling, RF Signal Processing, PAM/QAM Modulation

Languages: Python, C++, Java, TypeScript, JavaScript, R, Bash, Assembly, Dart

ML / Data: Feature Selection (HSIC-Lasso, Shapley Values), Expectation-Maximization, Computer Vision (YOLOv8, ResNet), TinyML, Statistical Modeling

Systems & Tools: gRPC, Yocto, Embedded Linux, FPGA (Xilinx), Git, Matlab, LaTeX, LTSpice, Jira, Confluence, Jenkins

Frameworks: Flutter, React, React-Native, Node.js

WORK EXPERIENCE

Nokia Solution and Networks Italia Spa., Jr. Software Engineer

Jul 2023 - Aug 2024

- Evaluated Rust as a memory-safe alternative to C/C++ for MSS-400 firmware (ownership model, type safety, C/C++ FFI); recommended deferring adoption until the package ecosystem matures.
- Automated DRBD/Pacemaker configuration (partition setup, resource-file generation, cluster bootstrapping) on the MSS-8 test system, replacing a manual, error-prone procedure.
- Deployed DRBD for real-time storage replication between active/standby control cards (MSS-E/MSS-HE cluster), syncing 500 MiB in under a minute; identified storage I/O as the bottleneck, not network throughput.
- Evaluated Pacemaker/Corosync (theoretical 99.999% uptime) and recommended against adoption for the two-node deployment — added complexity outweighed the benefit at that scale.
- Rewrote the indoor-unit discovery-protocol library's build system in Meson; benchmarked against the existing CMake build.
- Built a gRPC proof-of-concept for remote function invocation between client and server, validating the Capgemini-provided software stack.
- Built a Yocto BSP for a Xilinx FPGA, enabling embedded Linux on custom hardware.

Indian Institute of Science - Spectrum Lab, Research Assistant

Apr 2019 - Jun 2022

Medical image processing and estimation algorithms research under Prof. Chandra Sekhar Seelamantula; **7 peer-reviewed publications** (Nature Scientific Data, IEEE Access, IEEE TENCON, IEEE INDICON), Dice scores up to 0.97 across retinal, cardiac, and ultrasound imaging.

PROJECTS

Autonomous Driving and Secure Connected Vehicles

[GitHub Link: Click Here](#)

Event: *EESTEC Engineering Challenge 2026 organised by MESA*

Multi-sensor fusion pipeline (LiDAR, RADAR, camera, IMU, GPS) for opponent tracking and velocity estimation using YOLOv8m, on PoliMOVE data. **1st place** overall.

Semantic Communication Framework and Simulation with Sionna

[GitHub Link: Work in progress](#)

Thesis Project

gRPC-based semantic communication simulation using NVIDIA Sionna (PAM, 16-QAM physical layer). EM-based decoder: **95.75% accuracy at 0.36% feature transmission**; 128-dim variant: **96.13% accuracy, 8× lower decoder latency** (4.61s → 0.58s). Evaluated HSIC-Lasso and Shapley-value feature selection as alternatives to variance-based compression.

Trenord App with GeoFencing Features

[Report Link: Click here](#)

Course: *Design and Implementation of Mobile Applications*

Mobile app with geo-fencing for location-based notifications and ETA prediction.

Bluetooth-LE 6.0 Distance Estimation

[GitHub Link: N/A \(NDA\)](#)

Course: *Hardware Architecture for Embedded and Edge AI*

Benchmarked BLE 6.0 distance estimation (RSSI, PBR, RTT) against UWB DS-TWR using TinyML on Nordic nRF54L15 data.

Modular Hospital Ward Planner

[GitHub Link: Click here](#)

Course: *Computer Graphics*

Modular hospital ward planner in Vulkan: custom shaders, dynamic asset placement, multi-camera rendering.

PUBLICATIONS

RaPiD: A Raspberry Pi-based Optical Fundoscope

[Link: Click here](#)

Journal: *ISSS Journal of Micro and Smart Systems*

Automated Optic Disc Segmentation Using Basis-Splines-based Active Contour

[Link: Click here](#)

Journal: *IEEE Access*

Dice: 0.918–0.934 (4 datasets, 3000+ images).

Automated Segmentation of Common Carotid Artery in Ultrasound Images

[Link: Click here](#)

Journal: *IEEE Access*

Dice: 0.933 (lumen–intima), 0.918 (media–adventitia); +0.044 over prior SOTA.

Segmentation of Left Ventricle in Cardiac MR Images

[Link: Click here](#)

Archive: *ArXiv*

Dice: 0.873 (diastole), 0.770 (systole); 320 scans, M&Ms Challenge.

Chákṣu: A Glaucoma-specific Fundus Image Database

[Link: Click here](#)

Journal: *Scientific Data - Nature*

A Unified Approach for Automated Segmentation of Pupil and Iris in On-axis Images

[Link: Click here](#)

Journal: *Computer Methods and Programs in Biomedicine*

Dice: up to 0.971 (pupil), 0.905 (iris).

Automatic Segmentation of Fovea and Macula in Retinal Fundus Images

[Link: Click here](#)

Conference: *IEEE Region 10 (TENCON) 2022*

80.45% Dice, 99.66% fovea localization accuracy.

Automatic Optic Cup Segmentation Using Affine Snakes in Gradient Vector Field

[Link: Click here](#)

Conference: *IEEE INDICON 2023*